Exploring Quantum Recurrent Neural Networks in Sleep Medicine

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Abstract

Sleep medicine is devoted to the diagnosis and therapy of different sleep disorders with potential serious impacts over the health condition of individuals. The reference test to assess many of these disorders is the polysomnography (PSG). Clinical analysis of the PSG recording is expensive, requires substantial time and effort, and is prone to unintentional human misinterpretations, for which automation of this process has been extensively studied throughout the last years. The complexity of the task, on the other hand, offers an ideal scenario for the benchmarking of new developments in the field of machine learning with potential contributions to the task. This work examines the applicability of quantum machine learning for the diagnosis of sleep-disordered breathing, a common sleep disorder related to the repeated occurrence of involuntary respiratory pauses during sleep. By simulating density matrices and their state evolution, we attempt to emulate the behavior of a quantum recurrent neural network to study the feasibility of automatic classification of PSG sequences with presence or absence of respiratory events (apnea/hypopnea). A comparison with classical models is also made in order to evaluate the relative contribution of quantum artificial intelligence with respect to its classical counterparts.

Introduction

- Motivation
  - Polysomnography involves the recording of the patient's physiological signals during a prolonged period of sleep (6-8 hours). Visual inspection of the resulting recording is performed by the clinical specialist to detect complex patterns of neurophysiological activity to build a profile of the patient's sleep and diagnose any possible sleep-related disorders.
  - Due to the length and complexity of the task, as portrayed in figure 1, substantial time and effort is required, with potential occurrence of miss-scoring and annotation imprecision.

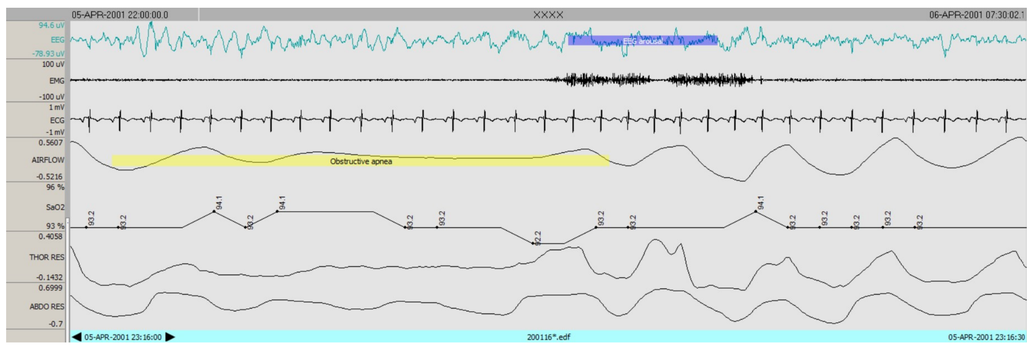


Figure 1. 30 s snapshot of a polysomnography showcasing microarousals and respiratory events

- Focus of the study
  - Through the application of artificial intelligence (AI), it is possible to greatly accelerate the classification of anomalous events, allowing professionals to invest less time and effort in the process activity.
  - Quantum artificial intelligence, a fusion of artificial intelligence and quantum computing, is an emerging discipline with great potential.
  - The complexity of the PSG analysis task offers an ideal scenario for the benchmarking of quantum artificial and machine learning methods in comparison to classical AI.
- Research goals
  - Sleep-disordered breathing is one of the most common sleep disorders related to the occurrence of apneas and hypopneas. The former is defined as a total obstruction in the airflow for a period of at least 10 seconds, while the latter represents a partial cessation.
  - The objective of the study is to automatically classify normal sleep periods with absence of apneas or hypopneas from those where these events are present. For this purpose we propose the use of a quantum machine learning model based on the concept of classical recurrent neural networks.

Datasets

- Data collection
  - The datasets have been constructed based on polysomnography recordings from different patients digitalized in the standard European Data Format (EDF). These files are processed using a windowing and normalization process in order to construct temporal segment instances.
- Characteristics
  - The datasets generated for the study have the following characteristics:
    - One single signal per instance, which represents the airflow of the patient.
    - 200–300 instances per dataset, where each instance is a time series of the airflow signal 30 s long.
    - A commons sampling frequency of 10 Hz is used, which makes the total number of values per sequence: $10 * 30 = 300$.

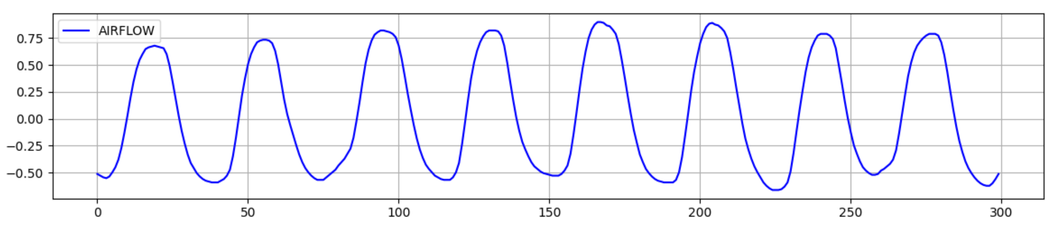


Figure 2. Graphic of normal instance with no event

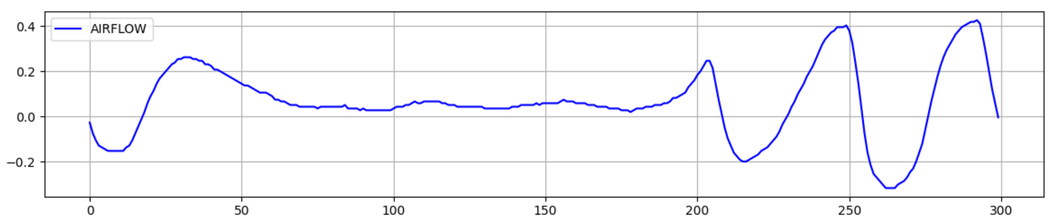


Figure 3. Graphic of event instance with an apnea being displayed

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Quantum Recurrent Neural Networks

- Recurrent Neural Networks
  - Recurrent neural networks are a type of neural network characterized by the use of memory, which allows them to remember past actions and make decisions based on them.
  - They are particularly suited for use with sequential data, as this memory makes continuous processing easier.

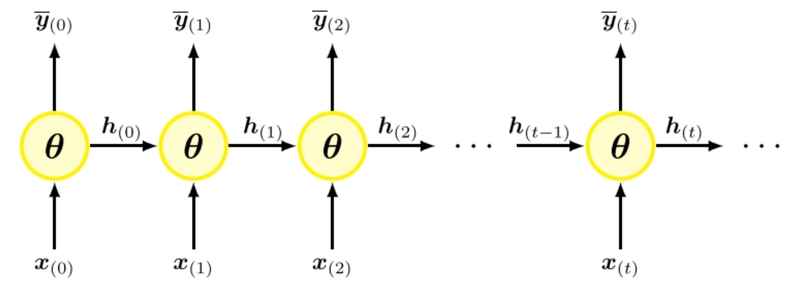


Figure 4. Diagram of a RNN

- In figure 4, the conceptual model of a the evolution of a recurrent neural networks through time presents the following sections:
  - θ: The parameters (weights) set of the model.
  - $x_{(t)}, y_{(t)}$: Inputs and their corresponding output labels at instant t
  - $h_{(t)}$: The hidden state at instant t. That is, the memory that is passed on to next iteration and influences future results.
- The output at timestep t is influenced by the corresponding input $x_{(t)}$ and the hidden state of the previous input $h_{(t-1)}$, making the impact of memory on decision-making effective.

- Developed quantum model
  - Quantum recurrent neural networks exploit basic properties of quantum mechanics, such as entanglement, to make a section of the resulting circuit act as memory. In this way, the values embedded in these qubits can influence subsequent outputs [4].

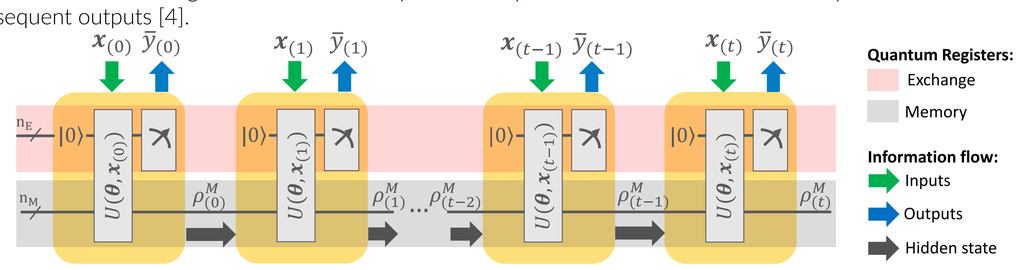


Figure 5. Diagram of a QRNN

- In figure 5, the model of the developed QRNN can be appreciated. The different parts that comprise it are as follows:
  - Exchange (E) and memory (M) registers. E consists of n_E qubits where input data are encoded. These qubits are measured and reset at the end of every timestep. We use the binary outputs from the first qubit measurements to classify the sample series. M consists of n_M qubits acting as memory and it is never measured.
  - ρ_M: The density matrix describing the memory state, which acts as the hidden state in the classical RNN and transmits information to the next timestep.
  - $U(x_{(t)}, \theta)$: The parameterised quantum operator encoding $x_{(t)}$ and evolving the quantum state. The parameter set θ is the same every timestep.

Obtained metrics

- Table 1 shows the results of different metrics for the execution of datasets of different sizes in the developed QRNN.
- These metrics have been constructed based on the tests score of a total of four executions per dataset.

Instances number	Accuracy	F1-score	Precision	Recall
150	0.76	0.44	1.0	0.30
200	0.89	0.89	0.87	0.90
300				

Conclusions

- At present, various tests are being carried out with the specifications indicated in the Datasets section.
- The results obtained demonstrate a clear ability to differentiate between the two classes, in many cases matching the level of classical models.
- In the future, more complex tests are expected to be carried out, with the main additions being:
  - Multi-class classification (differentiating between apnea class/hypopnea class, adding different sleep events such as microarousals, etc.).
  - Multi-signal classification (adding other signals that may be indicative of abnormalities).
  - Direct comparison, taking into account different metrics and aspects, with classical models.

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